James (Jimmy) Zhang
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EDUCATION

Northwestern University, Evanston, IL September 2023 – June 2027 (Anticipated)
Bachelor of Arts in Data Science, Mathematics, and Integrated Science Program (ISP)
GPA: 3.92/4.00
Coursework: Advanced Machine Learning, Deep Learning, Statistical Theory & Methods, Regression Analysis, Real Analysis, Linear Algebra, Mathematical Foundations of Neural Networks, Cell Biology

RESEARCH EXPERIENCE

Northwestern University Department of Physics and Astronomy June 2024 – Present
Undergraduate Researcher (Center for Network Dynamics)
Advisors: Prof. Adilson E. Motter, Dr. Thomas P. Wytock

- Investigate nonadditivity in *E. coli* gene expression that underlies observed phenotypic epistasis.
- Standardize text-based biological metadata for 3,500 experiments across 171 publications and organize related experimental conditions within perturbation networks using Python.
- Represent changes across 4,296 genes as differential expression vectors to characterize epistatic interactions through high-dimensional geometric relationships in the gene expression space.

SELECTED PROJECTS

Autoresearch for Dimension Reduction in Gene Expression April 2026 – June 2026

- Developed agentic AI workflow to identify optimal dimension reduction approach using Python.
- Tested 10 distinct methods over 58 autonomous experiments to maximize embedding quality for 946 brain regions and chromosome classification accuracy with multi-metric evaluation scheme.

QuickEDA: An Automated Tool for Data Science February 2026 – March 2026

- Developed web app to automatically conduct EDA and generate summary report for tabular data.
- Brainstormed project plan alongside generative AI and implemented application using Python.

Sentiment Analysis of Tweets October 2025 – December 2025

- Developed classification model to perform sentiment analysis on 40,000 Tweets using Python.
- Compared 16 deep learning architectures for natural language processing with baseline models from classical machine learning methods for multinomial classification.

Predicting Prices of U.S. Apartments January 2025 – March 2025

- Developed regression model to predict monthly rent prices for 100,000 U.S. apartments using R.
- Trained 1,359 candidate models across 2 feature engineering variations with cross-validated hyperparameter tuning to select and evaluate the best-performing model.

An Exploration of Natural Batter Swing Tendencies November 2024 – January 2025

- Analyzed relationships across 700,000 recorded swings from the 2024 MLB season to investigate batter tendencies in bat speed and swing length using R.
- Examined impact of game situations on swing tendencies and their relation to batting success.

PUBLICATIONS/PRESENTATIONS

Zhang, J, Wytock, T. P., and Motter, A. E. (2026). *The Geometry of Gene Expression Epistasis* [Manuscript submitted for publication].

Zhang, J. (2026). *Optimizing dimensionality reduction in gene expression using autoresearch* [Poster presentation]. Northwestern University Undergraduate Research and Arts Expo, Evanston, IL.

Last updated: October 2026

Zhang, J. (2025). *Understanding gene expression epistasis using complex networks and vector geometry* [Poster presentation]. Northwestern University Undergraduate Research and Arts Expo, Evanston, IL.

Zhang, J, Wytock, T. P., Motter, A. E. (2025). *The geometry of gene expression epistasis* [Conference presentation]. National Institute for Theory and Mathematics in Biology, Chicago, IL.

LEADERSHIP/TEACHING EXPERIENCE

Northwestern University Department of Mathematics September 2026 – Present
Undergraduate Teaching Assistant
- Lead weekly discussion sections, proctor and grade quizzes and exams, hold weekly office hours, and meet regularly with instructional team for single-variable integral calculus course.

Northwestern University Office of Undergraduate Research April 2026 – September 2026
Summer Research Peer Mentor
- Provided guidance to 8 grant awardees in navigating their first summer research experience.

Northwestern University Integrated Science Program September 2025 – Present
ISP Tutor
- Hold weekly 2-hour tutoring sessions in Math (primary) and Physics for first-year ISP students.

Northwestern University Department of Statistics and Data Science April 2025 – Present
Muntz Lassonde Teaching Assistant
- Support instructional faculty in introductory and advanced undergraduate data science courses.
- Hold weekly office hours, lead in-class review sessions, proctor examinations, grade assignments.

Northwestern University Marching Band March 2025 – Present
Drumline Captain
- Lead daily sectional rehearsals for 20 musicians and coordinate performance logistics with staff.

GRANTS AND HONORS

Phi Beta Kappa (Alpha of Illinois Chapter), *Northwestern University* 2026
- Elected as 1 of 25 juniors to the oldest academic honor society in the United States.

Barry M. Goldwater Scholarship, *Goldwater Foundation* 2026
- Received premier national undergraduate award for research in science, mathematics, and engineering, recognizing 400-500 students annually based on exceptional research potential.

Dean's List, *Northwestern University* 2023 – 2026 (8x)
- Awarded for earning a GPA of 3.7 or higher in coursework for a quarter.

Summer Undergraduate Research Grant (SURG), *Northwestern University* 2025, 2024
- Awarded $4,000 to conduct full-time research for 8 weeks based on submitted project proposal.

Fletcher URG Prize Winner, *Northwestern University* 2025
- Awarded for the most outstanding 2025 SURG project in Natural Sciences and Engineering.

Fletcher URG Prize Finalist, *Northwestern University* 2024

SKILLS

Computer: R (Tidyverse, Tidymodels, RStudio), Python (NumPy, Pandas, Matplotlib, Seaborn, SciPy, Sklearn, Keras, PyTorch, Flask), MATLAB, SQL, Git, LaTeX, Quarto
Technical: Agentic AI, Data Visualization, Data Wrangling, Deep Learning, Feature Engineering, Machine Learning, Natural Language Processing, Prompt Engineering, Statistical Inference
Language: English (Fluent), Chinese (Fluent), Latin (Proficient)
Activities: Marching Band/Drumline, Concert Band, Chess, Sports Analytics